General Mathematics — Exercise Sheet 41

Flux, surfaces and vector derivatives

Variant: Statements

Exercise 1 (A phase ellipse with a mixed term). Let $E > 0$ and let D be the region $2q^2+2qp+3p^2 \le 2E$ in the (q,p) plane. Find its area by a linear change of coordinates and compute $\oint_{\partial D} p\,\mathrm{d}q$ for anticlockwise traversal. The model equations are $\dot{q} = q + 3p$, $\dot{p} = -2q - p$. Use the same coordinate change to determine their direction and period on this ellipse, and the value of the integral for that motion. Use Lectures 22, 38 and 41 and the scalar oscillator solutions of Lecture 34.

Exercise 2 (Outward flux through a rectangle). Let $a, b > 0$, $D = [0,a] \times [0,b]$ and $\boldsymbol{F} = (x^3, y^3)$. Find outward flux by the planar divergence formula and by direct calculation on all four sides. State the outward normal on each side. Use Lecture 41 and proper integration from Lecture 26.

Exercise 3 (A graph and a reversed parameter order). Let S be $z = xy$ over $0 \le x, y \le 1$, and let $\boldsymbol{F} = (z, x, 2y)$. Compute upward flux using $\boldsymbol{r}(u,v) = (u, v, uv)$. Repeat with $\boldsymbol{R}(s,t) = (t, s, st)$, keeping the same upward physical orientation, and explain the sign needed in the second calculation. Use Lecture 41's vector area element and Lecture 13's cross product.

Exercise 4 (Area and flux of a spherical cap). Let $R > 0$, $-R \le h \le R$, $c \in \mathbb{R}$ and let S_h be the part of the sphere $x^2 + y^2 + z^2 = R^2$ with $z \ge h$, oriented outward. Find its area and flux for the constant field $\boldsymbol{F} = (0, 0, c)$. Include both endpoint caps. Use the angle convention and sphere vector area factor of Lectures 39 and 41.

Exercise 5 (Deriving the polar Laplacian). On a planar region with $r > 0$, choose a smooth local angular coordinate and write $f(x, y) = U(r, \varphi)$ with $U \in C^2$. Derive

$$\Delta f = U_{rr} + \frac{1}{r} U_r + \frac{1}{r^2} U_{\varphi\varphi}.$$

Find every radial harmonic function on an annulus $a < r < b$, $0 < a < b$. Decide which admit a radial harmonic continuation to the punctured disc $0 < r < b$ that then extends continuously to the origin. Also verify that $r^n \cos(n\varphi)$ is harmonic for every integer $n \geq 0$. Use the chain rule in Lecture 36 and Lecture 41's Cartesian definition.

Exercise 6 (A supplied spherical Laplacian and radial harmonics). As a given coordinate identity [C], for $r > 0$, $0 < \theta < \pi$ and a smooth local azimuth, a C^2 function $f = U(r, \theta, \varphi)$ satisfies

$$\Delta f = \frac{1}{r^2} \partial_r (r^2 U_r) + \frac{1}{r^2 \sin\theta} \partial_\theta (\sin\theta \, U_\theta) + \frac{1}{r^2 \sin^2\theta} U_{\varphi\varphi}.$$

Here θ is the polar angle from the positive z axis, as in Lecture 39. Source: NIST *Digital Library of Mathematical Functions*, equation 1.5.17 [C]. Use the supplied formula to find every radial harmonic function on a spherical shell $a < r < b$, $0 < a < b$, and every real power r^p that is harmonic. Then verify the power calculation directly from Cartesian derivatives, so that this last verification is independent of the supplied identity. Use Lectures 8, 36 and 41.